OXFORD AQA

INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname ______________________

Forename(s) ______________________

Candidate signature ______________________

I declare this is my own work.

INTERNATIONAL GCSE

BIOLOGY

Paper 2

Tuesday 4 June 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

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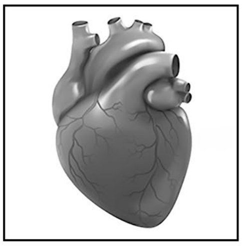
Answer **all** questions in the spaces provided.

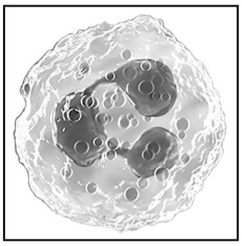
0 1 The circulatory system transports substances around the body in the blood.

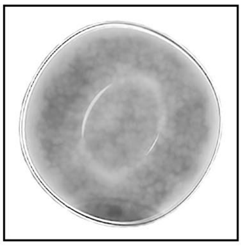
0 1 . 1 Draw **one** line from each structure in the circulatory system to the description of the structure.

[4 marks]

Structure in the circulatory system







Description

A small piece of a cell that starts the clotting process

Changes shape to ingest pathogens

Has a thick muscular wall for strong contraction

Has a wall one cell thick for diffusion

Has no nucleus so more space to carry oxygen

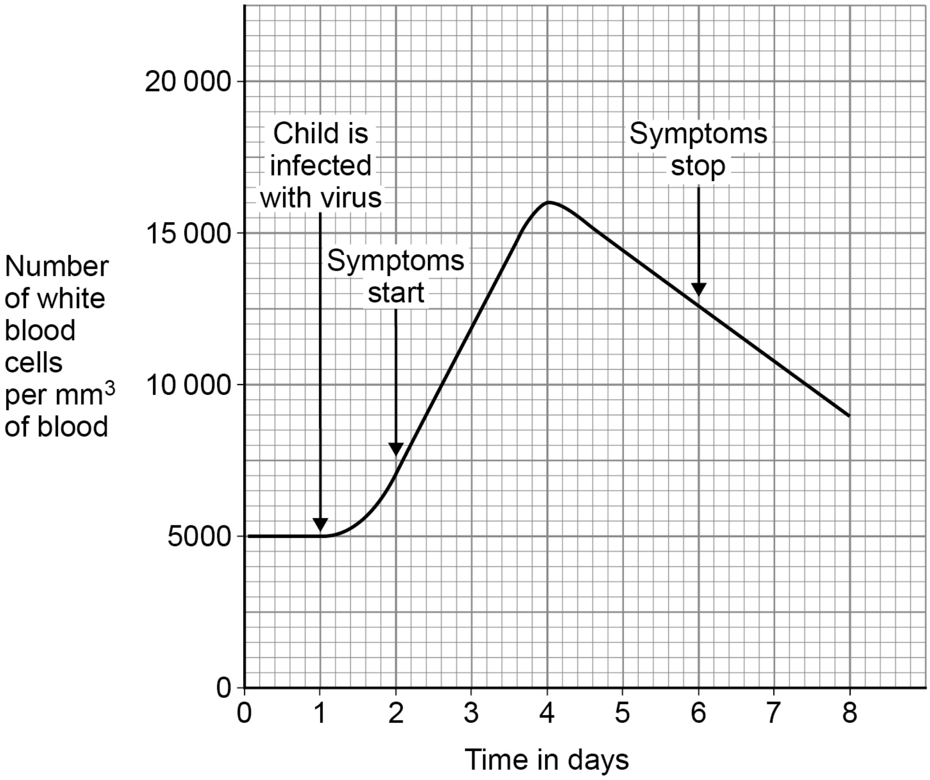
Turn over ►

A child became ill with an infection caused by a virus.

One symptom of the illness was fever.

Figure 1 shows how the number of white blood cells in the child's blood changed.

Figure 1



0 1 . 2 How many days did the child have symptoms?

[1 mark]

_______________________________ days

Do not write outside the box

0 1 . 3 Give **two** conclusions about the change in the number of white blood cells during the infection.

Use **Figure 1**.

[2 marks]

1 ____

____

2 ____

____

0 1 . 4 The white blood cells bind to antigens on the virus.

What type of chemical is an antigen?

[1 mark]

____

8

Turn over for the next question

Do not write outside the box

0 2 Multicellular organisms have different levels of organisation.

0 2 . 1 Draw **one** line from each level of organisation to the description of that level.

[3 marks]

Level of organisation	**Description**
Cell	A group of cells with a similar structure and function
Organ	A group of organs that perform a particular function
Organ system	A structure made of different tissues
Tissue	The smallest unit of living organisms

0 2 . 2 Write the structures from the box in order of organisation from the simplest to the most complex.

[3 marks]

blood	**circulatory system**	**heart**	**white blood cell**

Simplest ____________________

____

____

Most complex ____________________

Do not write outside the box

0 2 . 3 Cells become specialised to carry out particular functions in the body.

Name the process by which cells become specialised.

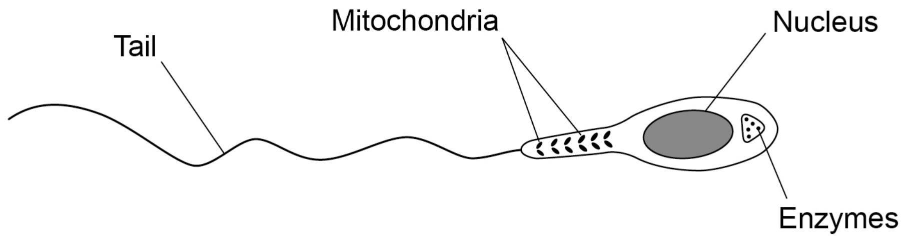
[1 mark]

____

0 2 . 4 A sperm cell is a male gamete found in humans.

Figure 2 shows a sperm cell.

Figure 2



Describe how **three** adaptations of the sperm cell are related to the function of the sperm cell.

Use **Figure 2**.

[3 marks]

1 ____

____

2 ____

____

3 ____

____

10

Turn over for the next question

There are no questions printed on this page

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ANSWER IN THE SPACES PROVIDED

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0 3 Different chemicals can be used to kill bacteria.

Two chemicals which kill bacteria are penicillin and the disinfectant bleach.

0 3 . 1 Give **one** situation where **penicillin** would be used to kill bacteria.

[1 mark]

0 3 . 2 Give **one** situation where **bleach** would be used to kill bacteria.

[1 mark]

Question 3 continues on the next page

Do not write outside the box

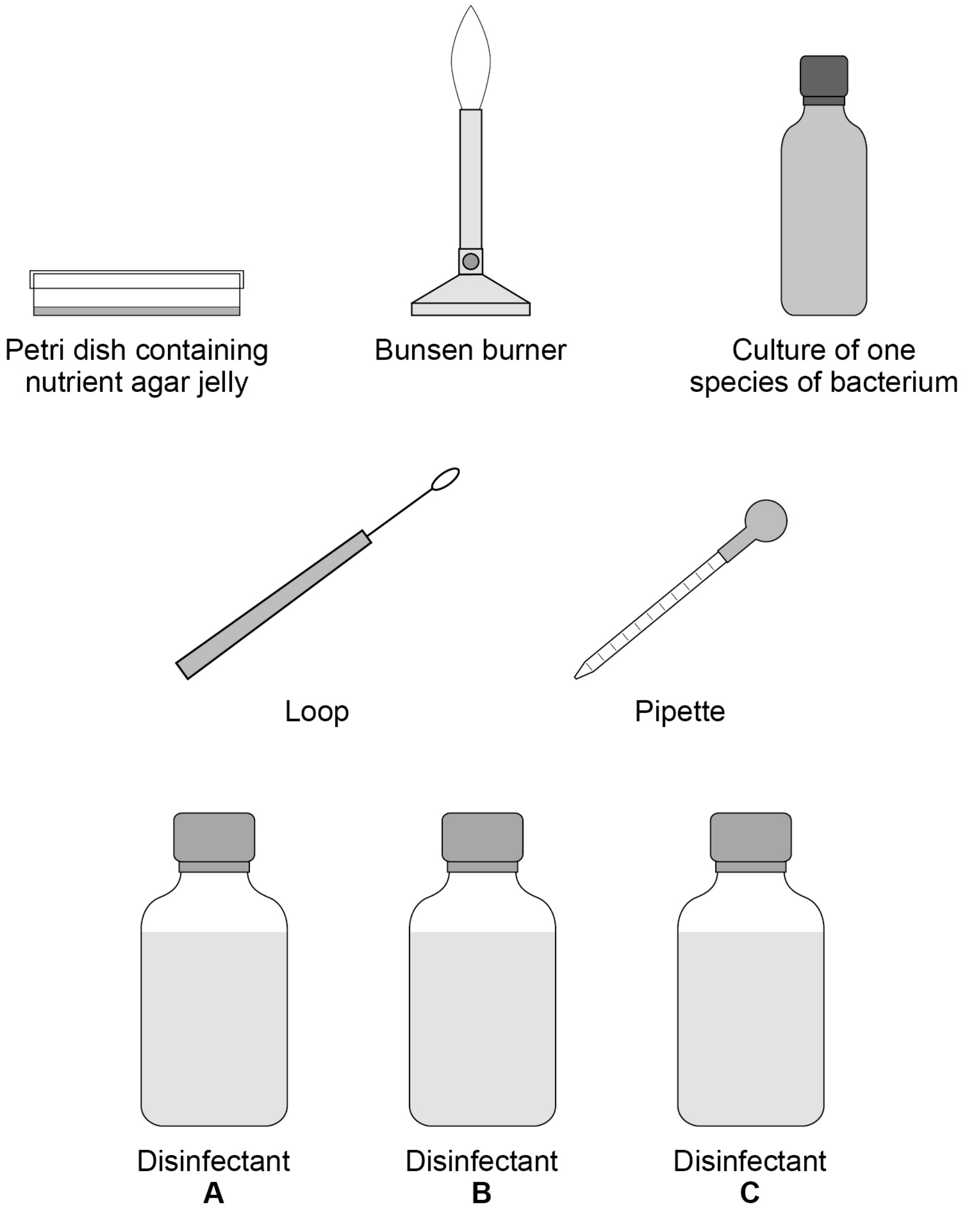
0 3 . 3 Three different disinfectants, **A**, **B** and **C**, also kill bacteria.

Describe a method to find out which of the disinfectants is the most effective.

[6 marks]

Figure 3 shows some of the equipment you can use.

Figure 3



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8

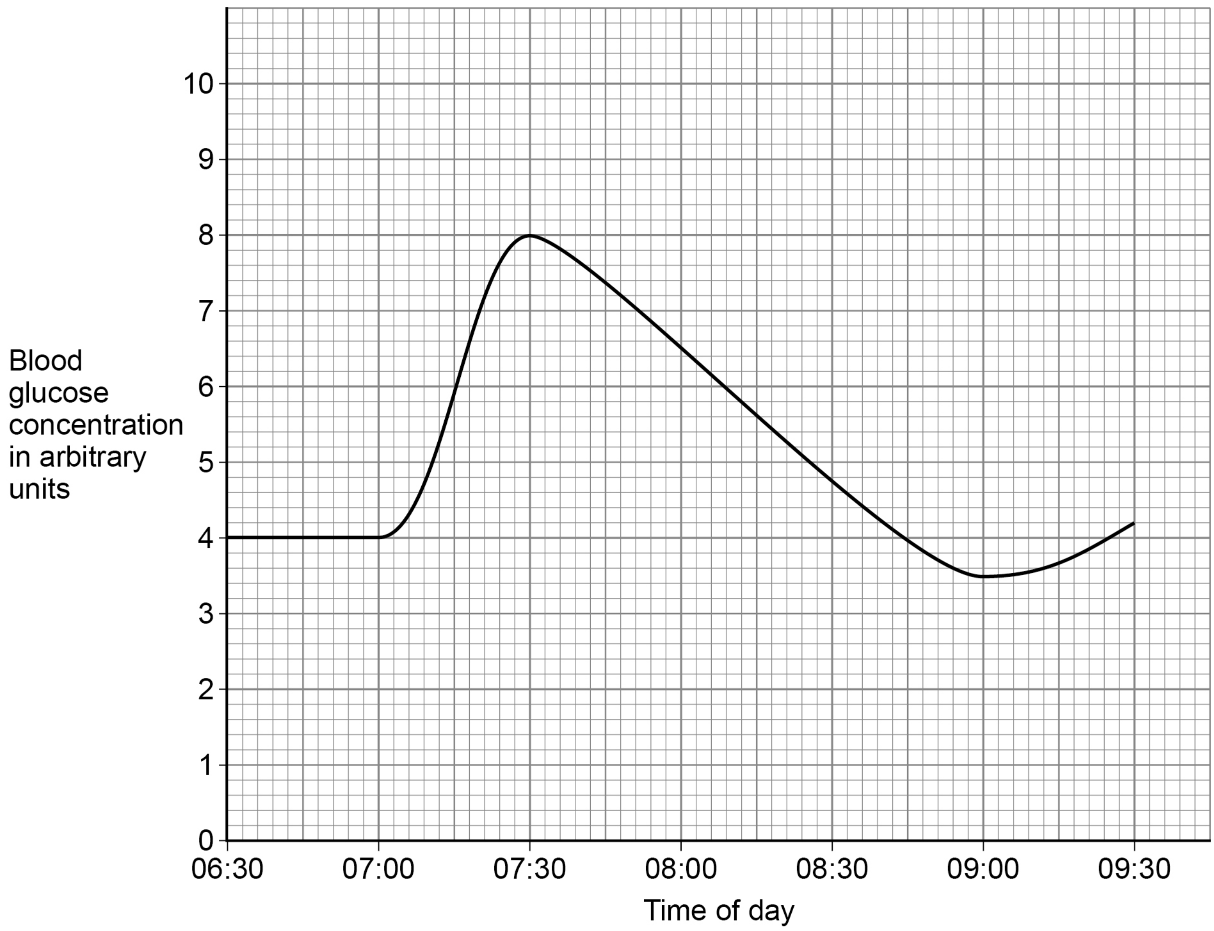
Turn over for the next question

0 4 Hormones control the concentration of glucose in the blood.

A person ate a meal at 07:00 but did not eat again until after 09:30.

Figure 4 shows the changes in blood glucose concentration.

Figure 4



0 4 . 1 What was the person's blood glucose concentration at 07:00?

Use **Figure 4**.

[1 mark]

____________________________ arbitrary units

Do not write outside the box

0 4 . 2 Describe the changes in blood glucose concentration between 07:30 and 09:30.

Include data from **Figure 4**.

[2 marks]

0 4 . 3 Explain how hormones control the changes in blood glucose concentration from 07:00 to 09:30 in **Figure 4**.

[6 marks]

9

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0 5 This question is about respiration.

0 5 . 1 Which organisms carry out aerobic respiration?

[1 mark]

Tick (✓) **one** box.

Animals only ☐

Plants only ☐

Both animals and plants ☐

0 5 . 2 When does aerobic respiration take place in a cell?

[1 mark]

Tick (✓) **one** box.

Day only ☐

Night only ☐

Both day and night ☐

0 5 . 3 Where does most aerobic respiration take place in a cell?

[1 mark]

Tick (✓) **one** box.

Chloroplasts ☐

Mitochondria ☐

Nucleus ☐

Do not write outside the box

0 5 . 4 A student swam 50 metres.

Explain why the rate of breathing and the depth of breathing of the student increased during the swim.

[4 marks]

Question 5 continues on the next page

Scientists monitored the lactic acid concentration in the blood of runners during a short and fast race.

Table 1 shows the results.

Table 1

Time in seconds	**Mean lactic acid concentration in mmol/dm^3**
0.0	1.2
5.0	2.8
7.0	6.9
9.0	8.3
11.0	12.5

0 5 . 5 Explain why the mean concentration of lactic acid increased during the race.

[2 marks]

Do not write outside the box

0 5 . 6 Calculate the percentage increase in the mean concentration of lactic acid by the end of the race.

Use the equation:

$$\text{percentage increase} = \frac{\text{increase in concentration from 0 to 11 seconds}}{\text{concentration at 0 seconds}} \times 100$$

Give your answer to 1 decimal place.

[4 marks]

percentage increase (1 decimal place) = ____________________ %

0 5 . 7 Describe what happens if lactic acid builds up in muscles.

[1 mark]

14

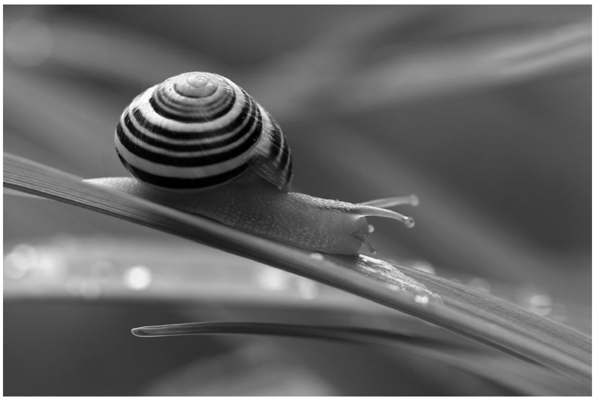
Turn over for the next question

Do not write outside the box

0 6 Individuals of the same species show variation in their characteristics.

Figure 5 shows a banded snail.

Figure 5



0 6 . 1 Variation may be continuous or discontinuous.

Which characteristic of banded snails is **continuous**?

[1 mark]

Tick (✓) **one** box.

Break in the shell ☐

Length of the body ☐

Number of bands on shell ☐

Do not write outside the box

0 6 . 2 Variation is caused by differences in:

- the genes an organism inherits
- the environment in which an organism lives
- a combination of both genes and environment.

Complete **Table 2** for the snail in **Figure 5**.

[3 marks]

Write **one** tick (✓) in **each** row.

Table 2

Characteristic	Cause of variation		
	Genes only	**Environment only**	**Both genes and environment**
Break in the shell			
Length of the body			
Number of bands on shell			

0 6 . 3 One cause of **genetic** variation within a species is sexual reproduction.

Give **one other** cause of genetic variation within a species.

[1 mark]

____

____

0 6 . 4 Giraffes have very long necks.

Describe how Lamarck's theory of evolution would explain the long necks of giraffes.

[2 marks]

____

____

____

____

Two different species of squirrel are found in different areas of the Grand Canyon in the United States of America.

The Kaibab squirrel lives in one small area that is high up in the Grand Canyon.

Figure 6 shows both species.

Figure 6

Abert's squirrel **Kaibab squirrel**

The Kaibab squirrel evolved from the Abert's squirrel over 10 000 years ago.

0 6 . 5 Explain how the Kaibab squirrel evolved to become a new species.

[6 marks]

13

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0 7 Exercise is essential to maintain good physical health.

A group of four students used a running machine to investigate the effect of exercise on heart rate.

This is the method used.

1. Measure the resting heart rate of each student and calculate the mean for the group.
2. Ask each student to exercise for 10 minutes at a speed of 3.0 kilometres per hour (km/hr) on the running machine.
3. Measure the heart rate of each student again and calculate the mean for the group.
4. Ask each student to rest for 20 minutes.
5. Repeat steps 2 to 4 at exercise speeds of 6.0, 9.0 and 12.0 km/hr.

0 7 . 1 The students controlled the length of time of exercise and the length of time of rest.

Give **two other** control variables the students should have used.

[2 marks]

1 ____

____

2 ____

____

The students used a stopwatch to count the heart beats in 30 seconds.

When they reset the stopwatch they noticed it was reading 1.5 seconds each time.

0 7 . 2 What type of error would this cause?

[1 mark]

____

Turn over ►

0 7 . 3 Suggest how the students could correct the error.

Do **not** refer to using a different stopwatch.

[1 mark]

0 7 . 4 Give **one** way the students could improve the investigation.

Do **not** refer to measurement of time.

[1 mark]

Table 3 shows the results.

Table 3

Exercise speed in km/hr	**Mean heart rate in beats per minute**
0.0 (at rest)	73
3.0	108
6.0	132
9.0	147
12.0	147

Do not write outside the box

0 7 . 5 Complete **Figure 7**.

You should:

- label the y-axis
- add the scale for the y-axis
- plot the data from **Table 3**
- draw a line of best fit.

0.0 km/hr has been completed for you.

[5 marks]

Figure 7

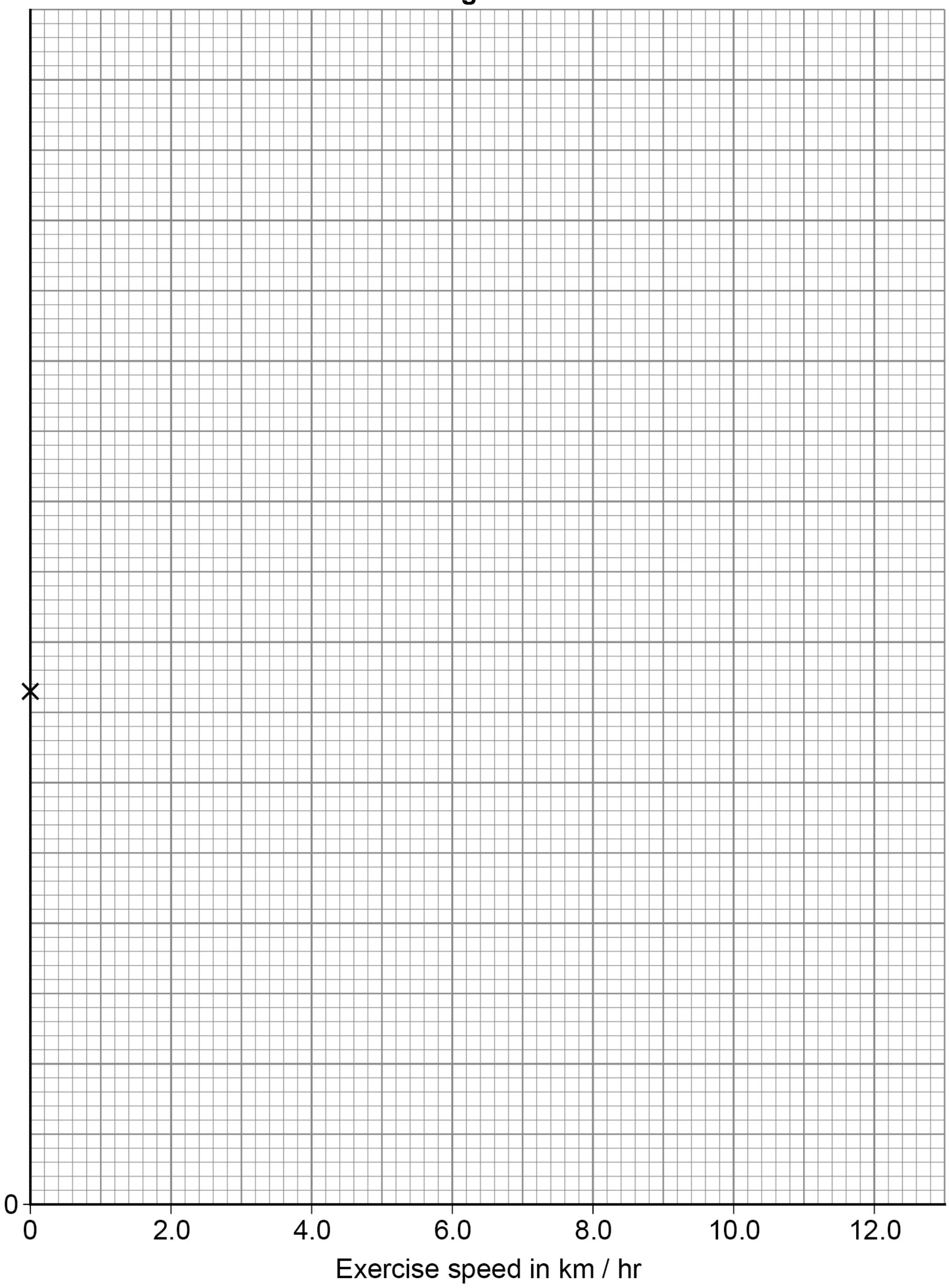

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Table 3 is repeated here.

Table 3

Exercise speed in km/hr	Mean heart rate in beats per minute
0.0 (at rest)	73
3.0	108
6.0	132
9.0	147
12.0	147

0 7 . 6 What conclusion can be made from the results?

Use data from **Table 3**.

[2 marks]

Question 7 continues on the next page

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Exercise is important for older people to help prevent them from falling.

An investigation was carried out in a care home for older women.

196 women aged 70 to 85 years were divided into three groups:

- **group 1** did extra exercise for 150 minutes per week
- **group 2** attended education talks about preventing falls
- **group 3** did not do extra exercise and did not attend education talks.

The number of falls for each group was recorded for 6 months.

Table 4 shows the results.

Table 4

Group	Number of women in group	Number of falls	% of women with falls	% reduction in falls compared to group 3
1. Extra exercise only	52	10	19.2	52.8
2. Education talks only	53	21	39.6	2.7
3. No exercise and no education talks	91	37	40.7	

0 7 . 7 Compare the effectiveness of exercise and education talks in preventing falls in older women.

Use data from **Table 4** in your answer.

[2 marks]

Do not write outside the box

0 7 . 8 A teacher compared the students' investigation with the care home investigation.

Both investigations looked at the effect of exercise on physical health.

The four students investigated the effect of exercise on heart rate.

The students' method is repeated below.

1. Measure the resting heart rate of each student and calculate the mean for the group.
2. Ask each student to exercise for 10 minutes at a speed of 3.0 kilometres per hour (km/hr) on the running machine.
3. Measure the heart rate of each student again and calculate the mean for the group.
4. Ask each student to rest for 20 minutes.
5. Repeat steps 2 to 4 at exercise speeds of 6.0, 9.0 and 12.0 km/hr.

Evaluate the **validity** of the students' investigation compared to the **validity** of the care home investigation.

[4 marks]

18

Turn over for the next question

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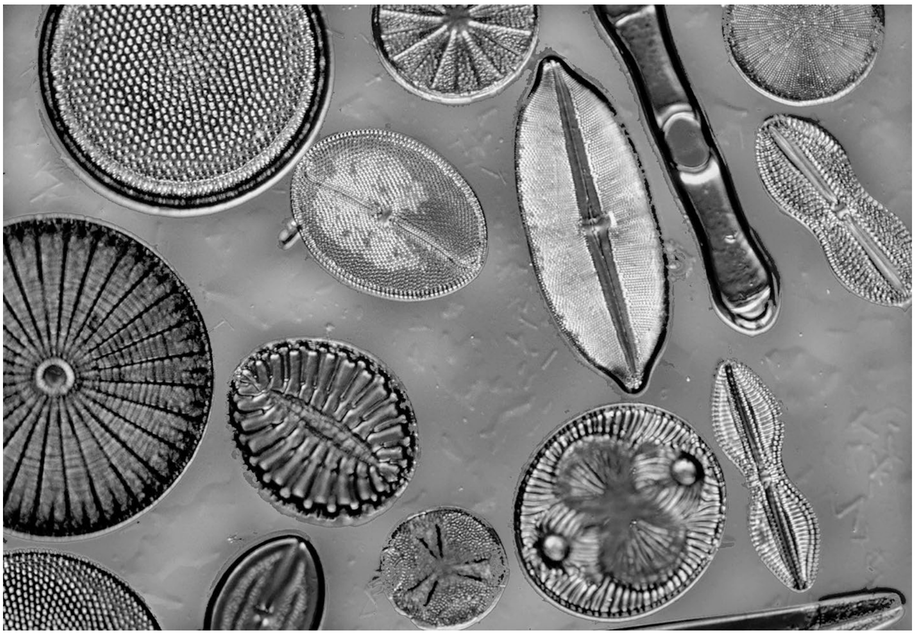
0 8 Diatoms are single-celled organisms that live in ponds, lakes and seas.

Figure 8 shows some diatoms.

Figure 8



0 8 . 1 One diatom is a sphere in shape.

The diameter of the diatom is 500 micrometres.

Calculate the surface area to volume ratio of the diatom.

You should use:

- surface area of a sphere = $4\pi r^2$
- volume of a sphere = $\frac{4}{3}\pi r^3$
- π = 3.14
- 1 metre = 1 000 000 micrometres

[6 marks]

Do not write outside the box

Surface area to volume ratio = ______________ : 1

0 8 . 2 Another diatom has a surface area to volume ratio of 9.0×10^5 : 1

A human has a surface area to volume ratio of 2 : 1

A diatom exchanges gases through its cell surface.

A human needs lungs to exchange gases.

Explain why the human needs a specialised surface to exchange gases but the diatom does not need a specialised surface.

[4 marks]

10

END OF QUESTIONS

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